

What Lies Within

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

There's something in the building. Can you retrieve the flag?

Hints

- There is data encoded somewhere... there might be an online decoder.
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Tools Used

- **file:** Identify image file type and properties
 - **strings:** Extract readable text from binary files
 - **exiftool:** Inspect image metadata and embedded text
 - **zsteg:** Detect steganography in PNG/BMP images
 - **base64:** Decode Base64-encoded data
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Complete Solution

Step 1: Download and Prepare

Download the file named `buildings.png` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file buildings.png
```

Expected output:

```
buildings.png: PNG image data, 657 x 438, 8-bit/color RGBA, non-interlaced
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings buildings.png | grep -i "picoCTF"
```

⚠ *If no results appear, proceed to the next step (decoding).*

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata:

```
exiftool buildings.png
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : buildings.png
Directory                   : .
File Size                    : 625 kB
File Modification Date/Time   : 2026:05:16 02:16:20+02:00
File Access Date/Time        : 2026:05:16 02:16:36+02:00
File Inode Change Date/Time   : 2026:05:16 02:16:31+02:00
File Permissions              : -rw-rw-rw-
File Type                    : PNG
File Type Extension          : png
MIME Type                    : image/png
Image Width                  : 657
Image Height                  : 438
Bit Depth                    : 8
Color Type                    : RGB with Alpha
Compression                  : Deflate/Inflate
Filter                       : Adaptive
Interlace                    : Noninterlaced
Image Size                   : 657x438
Megapixels                   : 0.288
```

The metadata itself is not suspicious.

Step 5: Analyze Steganography with zsteg

Use `zsteg` to inspect the PNG for hidden data in the LSB channels:

```
zsteg buildings.png
```

Output:

```
b1,r,lsb,xy      .. text: "^5>R5YZrG"
b1,rgb,lsb,xy    .. text: "picoCTF{<REDACTED>}"
b1,abgr,msb,xy   .. file: OpenPGP Secret Key
b2,b,lsb,xy      .. text: "XuH}p#8Iy="
b3,abgr,msb,xy   .. text: "t@Wp-_tH_v\r"
b4,r,lsb,xy      .. text: "fdD\""\\""\\" "
b4,r,msb,xy      .. text: "%Q#gpSv0c05"
b4,g,lsb,xy      .. text: "fDffffDD\""\\""
b4,g,msb,xy      .. text: "f\"ffff\""\\"DD"
```

b4,b,lsb,xy	.. text: "\"\$BDDDDf"
b4,b,msb,xy	.. text: "wwBDDDDfUU53w"
b4,rgb,msb,xy	.. text: "dUcv%F#A`"
b4,bgr,msb,xy	.. text: " V\"c7Ga4"
b4,abgr,msb,xy	.. text: "gOC_\$_@o"

The `b1,rgba,lsb,xy` channel contains repeated Base64-encoded text. Decode one instance to reveal the hidden message.

✅ Final Result

```
picoCTF{<REDACTED>}
```

🧠 Key Concepts

- **Steganography:** Data hidden in image pixels, often using the least significant bits (LSB).
 - **Metadata analysis:** Hidden fields like PNG tEXt chunks can reveal clues.
 - **Base64 encoding:** A common method to hide payloads in readable text.
 - **Tool chaining:** Combine `strings`, `exiftool`, and `zsteg` to uncover hidden content.
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🔗 Additional Resources

You can also use online tools to analyze steganography in images:

- [Steganography](#) – Online steganography analysis tool